

Capstone Project-MediAssist AI

Enterprise Healthcare Agentic AI Copilot

1. Overview

MediAssist AI is an enterprise healthcare Agentic AI copilot that combines LLMs, RAG, LangGraph, and MCP to provide intelligent, context-aware, and reliable healthcare assistance. It leverages specialized agents and multiple data sources to deliver accurate and validated responses.

Architecture Components

1. Users

Role

End users interacting with the system.

Input

- Queries
- Documents
- Images

Output

User requests sent to Streamlit Application.

2.Streamlit App

Role

Provides the user interface for interacting with the system.

Input

- User query
- Uploaded files
- Images

Output

Structured API request.

3. FastAPI Gateway

Role

Backend API layer responsible for request handling.

Input

Requests from Streamlit.

Output

Validated requests sent to LangGraph Supervisor.

4. LangGraph Supervisor

Role

Central orchestration engine.

Input

Validated request.

Output

Requests routed to specialized agents.

5. Planner Agent

Role

Creates execution plans.

Input

User query.

Output

Execution strategy.

6. Retriever Agent

Role

Retrieves relevant context from Vector Database.

Input

User query.

Output

Relevant document chunks.

7. MCP Agent

Role

Handles tool execution.

Input

Tool requests.

Output

Tool results.

8. Vision Agent

Role

Processes images and scanned documents.

Input

- Images
- PDFs
- Reports

Output

Extracted text and information.

9. Document Upload

Role

Receives documents for ingestion.

Input

PDFs and reports.

Output

Documents forwarded to chunking process.

10. Chunking**Role**

Splits documents into smaller chunks.

Input

Raw documents.

Output

Chunks.

11. Embedding Model**Role**

Converts chunks into vector representations.

Input

Document chunks.

Output

Embedding vectors.

12. Vector Database**Role**

Stores embeddings for semantic search.

Input

Embedding vectors.

Output

Relevant context for Retriever Agent.

13. PostgreSQL Database**Role**

Stores structured healthcare data.

Input

Patient records and transactional data.

Output

Structured information.

14. MCP Server

Role

Provides tools and database access.

Input

Requests from MCP Agent.

Output

Structured responses.

15. LLM Processing

Role

Core language model processing layer.

Input

- User query
- Retrieved context
- Tool outputs

Output

Draft response.

16. Observability (LangSmith)

Role

Monitors and traces system execution.

Input

Execution logs.

Output

Metrics and insights.

17. Reasoning Agent**Role**

Performs logical reasoning and inference.

Input

Draft response.

Output

Enhanced response.

18. Evaluator Agent**Role**

Validates generated responses.

Input

Response from Reasoning Agent.

Output

Validated response.

19. Response Agent**Role**

Applies formatting and output guardrails.

Input

Validated response.

Output

Final response delivered to the user.